Limitations of the AWS Learner Lab

Region restriction

All service access is limited to the **us-east-1** and **us-west-2** Regions unless mentioned otherwise in the service details below. If you load a service console page in another AWS Region you will see access error messages.

Service usage and other restrictions

The following services can be used. Specific limitations apply as documented. Any attempt to exceed a service limit, may result in immediate deactivation of the AWS account and all resources in the account may be immediately deleted. Services restrictions are subject to change.

Amazon API Gateway

- This service can assume the LabRole IAM role.

AWS App Mesh

Application Auto Scaling

- This service can assume the LabRole IAM role.

Amazon Athena

- This service can assume the LabRole IAM role.

Amazon Aurora

AWS Backup

AWS Batch

- This service can assume the LabRole IAM role.

AWS Certificate Manager (ACM)

AWS CloudFormation

- This service can assume the LabRole IAM role.

AWS CloudShell

- You can use Amazon Q suggestions in CloudShell if you switch from the default Bash to Z shell by running the zsh command. Once in the Z shell prompt, start typing a command to get suggestions. For example, type aws and then wait. It may offer a suggestion such as aws s3 cp s3://.... You can accept code completions by using the right arrow key on your keyboard. You can see alternate suggestions — if there are any available — before accepting a suggestion, by using the Tab key.

AWS CloudTrail

- This service can assume the LabRole IAM role.
- You can create a CloudTrail, but you cannot enable CloudWatch logging for the trail.

Amazon CloudWatch

AWS CodeCommit

- This service can assume the LabRole IAM role.

AWS CodeDeploy

- This service can assume the LabRole IAM role.

AWS Config

AWS Cost and Usage Report

AWS Cost Explorer

AWS DeepComposer

AWS DeepLens

AWS DeepRacer

- This service can assume the LabRole IAM role.

AWS Directory Service

Amazon DynamoDB

- This service can assume the LabRole IAM role.

Amazon EC2 Auto Scaling

- This service can assume the LabRole IAM role.
- Supported Instance types: nano, micro, small, medium, and large.
- Read the *Concurrently running instances limits* details documented in the EC2 service details below to be aware of further restrictions.
- *Recommendation*: size to your actual need to avoid using up your cost budget.

AWS Elastic Beanstalk

- This service can assume the LabRole IAM role.
- *To create an application*: choose **Create Application**, give it an application name, choose a platform, then choose **Configure more options**. Scroll down to the Security panel and choose **Edit**. For *Service role*, choose **LabRole**. If the environment is in the us-east-1 AWS Region, for *EC2 key pair,* choose **vockey** and for *IAM instance profile*, choose **LabInstanceProfile**. Choose **Save**, then choose **Create app**.

- Supported Instance types: nano, micro, small, medium, and large. If you attempt to launch a larger instance type, it will be terminated.

Amazon Elastic Block Store (EBS)

- Maximum volume size is 100GB
- PIOPs not supported

Amazon Elastic Compute Cloud (EC2)

- This service can assume the LabRole IAM role.
- Supported AMIs:
    - AMI available in us-east-1 or us-west-2. For example, Quick Start AMIs, My AMIs, and Community AMIs.
    - AWS Marketplace AMIs are not supported. AMIs such as MacOS that must launch as a dedicated instance or on a dedicated host are also not supported.
    - Recommendation: To launch an instance with a guest OS of ***Microsoft Windows***, ***Amazon Linux***, or one of many other popular Linux distributions, choose "Launch Instances", then choose from the ones available in the "Quick Start" tab.
- Supported Instance types: nano, micro, small, medium, and large.
- On-Demand instances only
- *Concurrently running instances limits* per supported region:
    - Maximum of 9 concurrently running EC2 instances, regardless of instance size. If you attempt to launch more, the excess instances will be terminated (and 9 will be left running).

Note: Service such as EMR, Elastic Beanstalk may also launch EC2 instances. The 9 concurrent running EC2 instances limit applies across all services that create instances visible in the EC2 console.

    - Maximum of 32 vCPU used by concurrently running instances, regardless of instance size or instance count. For example, t2.micro instances use 1 vCPU each, so you could run up to 32 of them in us-west-2 (but still only 9 of them in us-east-1 because of the other limitation listed above)

Note: The maximum 32 vCPU limit also applies across all services that create instances visible in the EC2 console.

    - **Caution**: Any attempt to have 20 or more concurrently running instances (regardless of size), will result in immediate deactivation of the AWS account and all resources in the account will be immediately deleted.
    - *Recommendation*: size to your actual need to avoid using up your cost budget.

- EBS volumes - sizes up to 100 GB and type must be General Purpose SSD (gp2, gp3) cold HDD (sc1), or standard.
- Key pairs - If you are creating an EC2 instance in any AWS Region other than us-east-1, the vockey key pair will not be available. In such cases, you should create a new key pair and download it when creating the EC2 instance. Then use the new key pair to connect to that instance.
- The EC2 Fleet feature is not supported in Learner Labs.
- A role named **LabRole** and an instance profile named **LabInstanceProfile** have been pre-created for you. You can attach the role (via the instance profile) to an EC2 instance when you want to access an EC2 instance (terminal in the browser) using AWS Systems Manager Session Manager. The role also grants permissions to any applications running on the instance to access many other AWS services from the instance.
- **Tips**:
    - When your session ends, the lab environment *may* place any running instances into a 'stopped' state.

If you have enabled "Stop protection" on any instances, this protection will be removed when the session ends and the instance will still be stopped. Stop protection will *not* be re-enabled automatically when you later start a new session.

    - When you start a new session, the lab environment will start all instances that were previously stopped by you or stopped by the lab environment when the lab session ended.
    - Instances that have been stopped and started again will be assigned a new IPv4 public IP address unless you have an elastic IP address associated with the instance.
- **Recommendations**:
    - To preserve your lab budget, stop any running EC2 instances before you are done using the account for the day (and terminate them if not longer needed).
    - Be aware of all instances you keep in the account between sessions because they will run (and cut into your budget) when you start the lab again unless you remember to turn stop them manually after starting the lab.

Amazon Elastic Container Registry (ECR)

- The LabRole IAM role has read-only access to this service and as a console user you have write access to this service.

Amazon Elastic Container Service (ECS)

- Supported Instance types: nano, micro, small, medium, and large.

- Tips:
    - **When creating a cluster**: To avoid permissions errors (cannot create role), if attempting to create a cluster that uses EC2 instances for the infrastructure (instead of Fargate), create an auto scaling group using the EC2 console first, then choose to use that existing auto scaling group when creating the ECS cluster.
        - If you see a message "*the ECS service linked role could not be assumed*" choose the back button and then try again. This sometimes happens if the service linked role does not yet exist in your account.
    - **When creating a task definition**: To avoid permissions errors, be sure to set *LabRole* as the task role and task execution role.

Amazon Elastic File System (EFS)

- This service can assume the LabRole IAM role.

Amazon Elastic Kubernetes Service (EKS)

- This service can assume the IAM Roles having identifier *LabEksClusterRole* created for *Cluster* and *Node*.
- Supported Instance types: nano, micro, small, medium, and large.

Elastic Load Balancing (ELB)

- This service can assume the LabRole IAM role.

Amazon Elastic MapReduce (EMR)

- This service can assume the LabRole IAM role.
- Supported Instance types: nano, micro, small, medium, and large. If you attempt to launch a larger instance type, it will be terminated.
- **Tip**: If you have any trouble successfully launching a cluster, try using the m4.large instance type. *Example* EMR cluster configuration details:
    - *Cluster configuration*: Instance groups, Primary: **m4.large**, Core: **m4.large**, Task -1: **m4.large**
    - *Provisioning configuration*: Core: 1 instance and Task -1: 1 instance
    - *Security configuration*: EC2 key pair: **vockey**
    - *IAM roles*: *Amazon EMR service role*: choose existing: **EMR_DefaultRole**, *EC2 instance profile for Amazon EMR*: choose existing: **EMR_EC2_DefaultRole**
- **Note:** If you encounter the issue with S3 bucket creation as part of cluster creation:

    - Expand the section - **Cluster logs** and **Uncheck** the value for *Publish cluster-specific logs to Amazon S3*, and proceed with cluster creation. The logs will not be stored in s3 bucket, but cluster will get created.

- Maximum of 32 vCPU used by concurrently running EC2 instances in an AWS Region. Note that you are also limited to launching no more than nine (9) instances (of any size) in a Region at once.

Note: *An EMR cluster will not continue to work if your session ends*. In Learner Labs, session end causes the EC2 instances that the EMR cluster uses to be stopped, and stopping an EMR cluster is not supported (by AWS). *Recommendation*: write EMR job results to S3 if you need to preserve your results, before you end your current Learner Labs session, then read the results back into a new EMR cluster as needed when you start your next Learner Labs session.

Amazon ElastiCache

Amazon EventBridge

- This service can assume the LabRole IAM role.
- Use LabRole if creating an EventBridge Pipe.

AWS Fargate

- This service can assume the LabRole IAM role.

AWS Glue

- This service can assume the LabRole IAM role.

Note: The following limitations apply to the AWS Glue ETL job configuration

    - Allowed Worker type: G.1X, Standard
    - Maximum number of workers: 10
    - Maximum Concurrency: 1

AWS Glue DataBrew

- This service can assume the LabRole IAM role.

Amazon GuardDuty

AWS Health

AWS Identity and Access Management (IAM)

- Extremely limited access. You cannot create users or groups. You cannot create roles, except that you can create service-linked roles.
- Service role creation is generally permitted. If the service needs to create a role for you, you may need to retry role creation if it fails the first time.

- A role named **LabRole** has been pre-created for you. This role is designed to be used when you want to attach a role to a resource in an AWS service. It grants many AWS services access to other AWS services and has permissions very similar to the permissions you have as a user in the console.
    - Example use: attach the LabRole via the instance profile named **LabInstanceProfile** to an EC2 instance for terminal in the browser access to an EC2 instance guest OS using AWS Systems Manager Session Manager.
    - Another example: Attach the LabRole to a Lambda function so that the Lambda function can access S3, CloudWatch, RDS, or some other service.
    - Another example: Attach the LabRole to a SageMaker notebook instance so that the instance can access files in an S3 bucket.

Amazon Inspector

AWS IoT 1-Click

AWS IoT Core

- This service can assume the LabRole IAM role.

AWS IoT Greengrass

AWS Key Management Service (KMS)

- This service can assume the LabRole IAM role.

Amazon Kinesis

- If attempting to create a *Kineses Data Analytics Studio notebook*, choose "Create with custom settings" and then choose **LabRole** in the IAM settings area.
- If attempting to create a *Kinesis Delivery Stream*, choose "Advance settings" and then choose to use the existing **LabRole**.

AWS Lambda

- **Tip**: Attach the existing **LabRole** to any function that you create if that function will need permissions to interact with other AWS services.
- A maximum of 10 concurrent running Lambda execution environment instances are permitted. Reference the documentation to understand how concurrency is calculated.

Amazon Machine Learning (Amazon ML)

AWS Marketplace Subscriptions

- Extremely limited read-only access.

AWS Mobile Hub

AWS OpsWorks

Amazon Q Developer

- Amazon Q Developer offers many features, including inline code suggestions that can be used in a variety of code editors including the AWS Lambda code editor. For a listing of IDEs that support generating inline suggestions, see the AWS Documentation.

Note: Some IDEs refer to the auto-suggestion feature under the name CodeWhisperer. CodeWhisperer features are moving to Amazon Q Developer.

Amazon Redshift

- This service can assume the LabRole IAM role.
- Supported instance type: ra3.large
- Supported cluster size: maximum two instances

Amazon Rekognition

- This service can assume the LabRole IAM role.

Note: Service usage is subject to the following limits, which are sufficient for exploring service features or creating sample apps for educational purposes.

    - Rekognition Detected Face Limit - 1000
    - Rekognition Detected Label Limit - 1000
    - Rekognition Max Inference Unit - 1
    - Rekognition InService Inference Unit -1

Amazon Relational Database Service (RDS)

- This service can assume the LabRole IAM role.
- Supported database engines: Amazon Aurora (Provisioned), Oracle, Microsoft SQL, MySQL, PostgreSQL and MariaDB. Note: if you are creating an RDS instance using a CloudFormation template, be sure to specify the engine type using lower-case letters.
- Supported instance types: nano, micro, small, and medium (Tip: choose *Burstable classes* to find these).
- Supported storage types: EBS volumes - size up to 100 GB and type General Purpose SSD (gp2). PIOPS storage types are not supported.
- On-Demand DB instance class types only.
- ***Enhanced monitoring is not supported*** (you must *uncheck* this default setting in the *Additional configuration / Monitoring* panel).

- **Tip**: to preserve your lab budget, stop any running RDS instances before you are done using the account for the day (or terminate them if not longer needed).
- **Caution**: When a lab sessions ends, the lab environment may not stop an RDS instance or cluster that you leave running. Also, even if you *do* stop an RDS instance, if you leave it stopped for seven days, *AWS will start it again automatically*, which will increase the cost impact.

AWS Resource Groups & Tag Editor

- This service can assume the LabRole IAM role.

Amazon Route 53

- You cannot register a domain.

Amazon SageMaker

- This service can assume the LabRole IAM role.
- Supported SageMaker instance types: ml.t3.medium, ml.t3.large, ml.t3.xlarge, ml.m5.large, ml.m5.xlarge, ml.c5.large, ml.c5.xlarge only.
- Maximum Sagemaker Notebooks: 2
- Maximum Sagemaker Apps: 2
- To **create** a **SageMaker Domain**:
    - Choose **Domains** and then choose **Create domain**.
    - Choose **Set up for organizations** and choose **Set up**.
        - Step 1: Set up Domain details & users
            - Give the domain a name such as myDomain.
            - Keep the **Login through IAM** default.
            - Leave *Who will use Sagemaker* blank.
            - Choose **Next**.
        - Step 2: Configure roles and ML activities
            - Choose **Use an existing role**.
            - Set the *Default execution role* to **LabRole**.
            - Choose **Next**.
        - Step 3: Configure Applications

Note: Ignore and close any message about servicequotas:RequestServiceQuotaIncrease permissions issues.

        - In the *StageMaker Studio* panel, choose **SageMaker Studio - New**.
        - In the *JupyterLab* panel
            - Choose **Enable idle shutdown** and set it to 60 minutes.
            - Choose **Allow users to set custom idle shutdown time** and set the Maximum to 600.
        - In the *Canvas* panel
            - Choose **Configure Canvas**.
            - Scroll down to **Canvas Ready-to-use models configuration** and choose **Use an existing execution role**.
            - For **Execution role name**, choose **Enter a custom IAM role ARN**.
            - For Custom IAM role ARN, paste in the ARN of LabRole. It will be in the formation arn:aws:iam::ACCOUNT_ID:role/LabRole where ACCOUNT_ID is your actual Account ID.
        - In the *CodeEditor* panel:
            - Choose **Enable idle shutdown** and set it to 60 minutes.
            - Choose **Allow users to set custom idle shutdown time** and set the Maximum to 600.
    - Step 4: Customize Studio UI - Optional
        - Scroll to the bottom and choose **Next**.
    - Step 5: Set up network settings
        - Choose either VPC only or *Public internet access* are both options.
        - Example:
            - Choose **VPC Only**.
            - For VPC for Studio to use: Choose the Default VPC.
            - Choose the **VPC Console** link. A new browser tab opens the VPC console. In the *Resource map* tab, make note of the subsets that are public. Public subnets are

the ones that are connected to a route table that routes to an "igw-..." (internet gateway) network connection.

- Back in the SageMaker console, choose at least two of the identified public subnets.
- Choose the default security group.
- Choose **Next**.

- Step 6: Configure storage
  - Scroll to the bottom and choose **Next**.
- Step 7: Review and create
  - Scroll to the bottom and choose **Submit**.

- Wait for the Domain to be created. It typically takes 5 to 8 minutes to complete. Refresh the browser tab occasionally to more quickly know when it has completed.

- To create a SageMaker **user profile**:
  - In the list of SageMaker domains, choose the name link of the domain you created.
  - In the User profiles tab, choose **Add user**.
  - In *General settings*, for *Execution role* choose **LabRole** and then choose **Next**.
  - In *Step 2*: Choose **Next**.
  - In *Step 3*: Choose **Next**.
  - In *Step 4*: Keep *Inherit settings from domain* selected and choose **Next**.
  - In Step 5: Scroll to the bottom and choose **Submit**.
- To create a Code Editor space, so that you can use **Visual Studio Code Open Source**:
  - Choose **Studio**.
  - In the *Get started* panel, verify the user profile you created is selected and choose **Open Studio**.
  - Choose **Skip Tour** for now then from the **Applications** panel choose **Code Editor**.
  - Choose the **Create Code Editor space** button in the top right corner.
  - Give it a name such as mySpace and then choose **Create space**.
  - Verify the space settings before running it. The defaults, such as the default ml.t3.medium instance type are supported.

    - Choose **Run space**.
    - Once the space has started choose Open Code Editor.

A new browser tab opens, displaying an IDE. Refer to the AWS Documentation to learn more about Code Editor in SageMaker Studio.

- There is limited support for **SageMaker Studio** features. Some SageMaker JumpStart projects require more access permissions than we can grant in Learner Labs.
    - To use SageMaker Studio, first create a SageMaker domain and user profile (steps documented above).
    - Once you have a domain and user profile, choose **Studio**. Then, from **Applications** list in the top left corner, choose an application.
        - *For example*, choose **JupyterLab** and then choose **Create JupyterLab space**. Give the space a name and choose **Create space**. Then choose **Run space**. Wait for the JupyterLab space to start, then choose **Open JupyterLab**. Once in the JupyterLab UI can you claunch a notebook, console, or other type of resource.
    - If you have a running *studio space*, it will appear as a running *App* in the SageMaker Studio console.
- There is limited support for **SageMaker Canvas** features. There are many SageMaker Canvas models (including many Ready-to-use models) that are not supported in Learner Labs. For example, if a model is "powered by" an AWS service that is not supported in Learner Labs, then the model will not run in Learner Labs.
    - To use SageMaker Canvas, first create a SageMaker domain and user profile (see the steps above).
    - Once you have a domain and user profile, from the *User profiles* list in the row with the profile you want to use, choose **Launch** > **Canvas**. The SageMaker Canvas console appears.
- You can create **SageMaker Notebook instances**.
- **Tips to preserve your budget**:
    - Choose the **SageMaker dashboard** link to view recent activity including running jobs, models, or instances. Stop or delete anything that is running and that you no longer need.
    - When your session ends, the lab environment *may* place running SageMaker notebook instances into a 'stopped' state. Stopped SageMaker notebook instances will *not* be automatically restarted when you start a new session.

    - When using SageMaker Canvas or SageMaker Studio, logout of the **session** when you are done working with it. *Consider deleting SageMaker Canvas and SageMaker Studio apps that are no longer needed*.

AWS Secrets Manager

- This service can assume the LabRole IAM role.

AWS Security Hub

AWS Security Token Service (STS)

AWS Serverless Application Repository (SAR)

AWS Service Catalog

- This service can assume the LabRole IAM role.

Amazon Simple Notification Service (SNS)

- This service can assume the LabRole IAM role.

Amazon Simple Queue Service (SQS)

- This service can assume the LabRole IAM role.

Amazon Simple Storage Service (S3)

- This service can assume the LabRole IAM role.

Amazon Simple Storage Service Glacier (S3 Glacier)

- You cannot create a vault lock

Amazon Simple Workflow Service (SWF)

AWS Step Functions

AWS Systems Manager (SSM)

- A role named **LabRole** and an instance profile named **LabInstanceProfile** have been pre-created for you. You can attach the role (via the instance profile) to an EC2 instance when you want to access an EC2 instance (terminal in the browser) using AWS Systems Manager Session Manager.

Amazon Textract

AWS Trusted Advisor

Amazon Virtual Private Cloud (Amazon VPC)

AWS WAF - Web Application Firewall

AWS Well-Architected Tool